TECHNICAL DILIGENCE REVIEW - RECORD 007

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-007

1. Scope of the component and licence review

The instruction behind the report at TDR-2025-24 was narrow: examine the third-party component inventory, licence exposure and contribution provenance of the codebase and report by 2025-11-20. As instructed, Calder Rowe Engineering Advisory confined the present report to the third-party component inventory, licence exposure and contribution provenance of the codebase; three adjacent questions raised during the work are listed as follow-ups rather than answered here. This report would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. The pack this report proposes is deliberately short, because a longer one would be produced and not read. The company reviewed a draft of the present report for factual accuracy and its corrections are reflected without altering any conclusion.

5 sites contributed evidence to this report, and one further site was in acceptance testing and was excluded. M. Quillan chose the sample for this report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. The present report carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. Nothing in report TDR-2025-24 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. M. Quillan has stated the limitations at the point they bite rather than collecting them into a disclaimer.

M. Quillan was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. The present report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. Where the present report cannot conclude, it says what evidence would let it conclude. The open items carried forward from the present report are set out below; a reader should treat the committed dates as intentions rather than facts. M. Quillan has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

This report is limited by log retention: the telemetry window is ninety days, so events before 2026-02-20 could not be examined directly. Where report TDR-2025-24 reports a supplier's test result, the reviewer read the certificate and did not witness the test. Open items in the present report are stated with what would close each one, so a post-close plan can be costed against them. This report preserves 5 unresolved questions rather than resolving them by assumption. Calder Rowe Engineering Advisory has reported the position as at 2025-11-20 and has not attempted to predict where it will be at completion.

Grading in report TDR-2025-24 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. The report at TDR-2025-24 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. The population for the report at TDR-2025-24 was the service inventory as at 2025-11-20; 18 entries were examined and the rest were reviewed only at inventory level. Sampling in TDR-2025-24 covered 18 of the 34 catalogued services and 5 of the 18 live sites. The numbering in the present report follows the order of the work rather than the order of importance.

Minutes of every the present report interview were returned to the participant, and three came back with substantive corrections. The reviewer found the interview evidence for this report candid, and in two cases more candid than the written record. 4 items in this report are post-close engineering work rather than transaction issues, and the reviewer has costed them at $1,030,000. This report recommends that 1 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. The report at TDR-2025-24 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Schedule T - Third-party component and licence inventory (TDR-2025-24)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically	approved 2023-	Costmap Fusion

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
				linked by the fusion pipeline	08-04	
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding across edge and cloud	approved 2024-04-08	Config Distribution
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip detector, not shipped	not approved	Wheel Odometry Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

2. Component inventory and licence classification

The reviewer's note on pgbouncer-managed is that attribution present. Meeting the terms on pgbouncer-managed means to share alike any derived asset it distributes. Public registry, signature verified is the provenance recorded for pgbouncer-managed. pgbouncer-managed was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Public registry, signature verified is the provenance recorded for protobuf-runtime. Protobuf-runtime shows an approval status of approved 2024-04-08 against the company's own policy. The terms applying to protobuf-runtime are BSD-3-Clause; the copyleft classification is no. protobuf-runtime is the reason one section of TDR-2025-24 concludes narrowly rather than generally.

Safety-controller-toolchain entered the codebase by this route: supplied by the safety controller vendor under a written licence. The inventory entry OSS-019 is safety-controller-toolchain 3.0.4, safety controller build toolchain. Commercial, development use only governs safety-controller-toolchain, which the inventory marks as no copyleft. safety-controller-toolchain is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Pager-provider-sdk sits inside Alarm Router and is therefore in the shipped product rather than in tooling. The terms applying to pager-provider-sdk are commercial, subscription; the copyleft classification is no. Licence text and the actual usage of pager-provider-sdk were read together, and the older application interface is deprecated by the provider with an end date in 2027. pager-provider-sdk was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The distribution question for cell-model-lib turns on the fact that it is used battery state estimation. In addition to its terms, cell-model-lib brings a commercial supplier whose support statement expires inside the plan period. The supply route for cell-model-lib is public registry, signature verified. cell-model-lib was tested rather than accepted, and the test is described in the working papers behind TDR-2025-24.

OSS-025 records wheel-slip-model 0.3.1 against Wheel Odometry Filter, prototype slip detector, not shipped. To keep the dynamic-linking boundary intact is what wheel-slip-model requires in practice. One point about wheel-slip-model is not resolved: a network-copyleft component in a prototype; the reviewer flagged the shipping risk before validation completes. wheel-slip-model rests on a single source, and the reviewer has said so rather than implying corroboration.

What the company must actually do about kafka-client is to publish the modified files it distributes. Tracing kafka-client to its source gives public registry, signature verified. Policy compliance for kafka-client rests on its approval entry, which is approved 2024-02-27. kafka-client is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Public registry, signature verified is the provenance recorded for pointcloud-utils. The operational obligation attaching to pointcloud-utils is to keep the dynamic-linking boundary intact. One point about pointcloud-utils is not resolved: file-level copyleft; three modified files must be published and the reviewer could not evidence publication. pointcloud-utils has no single accountable owner, and that is the finding the reviewer would press hardest.

The distribution question for scanmatch-legacy turns on the fact that it is used linked into the localization fallback path. On provenance, scanmatch-legacy shows vendored source copied from a project archive in 2022. The non-licence risk on scanmatch-legacy is a single-maintainer dependency with no succession plan. scanmatch-legacy appears twice in the working papers for TDR-2025-24 because the first answer was corrected after a follow-up request.

The operational obligation attaching to grpc-transport is to share alike any derived asset it distributes. Grpc-transport was obtained as follows: public registry, signature verified. Licence text and the actual usage of grpc-transport were read together, and no notice obligations beyond attribution. grpc-transport is one of the entries where TDR-2025-24 relies on the company's own measurement rather than an independent one.

Schedule C - Producer and consumer interface contracts examined (TDR-2025-24)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	1 in the reviewed period
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	none
Costmap Fusion	motion planner	published schema	v3.9	not automated	none
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	2 in the reviewed period
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	none
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the pipeline	2 in the reviewed period
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	1 in the reviewed period
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	none
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	2 in the reviewed period
Charge Scheduler	fleet health monitor	request-response interface	v1.8	not automated	1 in the reviewed period
Safety Controller	drive power contactors	published schema	v2.5	advisory only	1 in the reviewed period
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	none
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	none
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	1 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	none
Lidar Driver	localization engine	published schema	v3.1	advisory only	1 in the reviewed period
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	1 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	2 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	1 in the reviewed period
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	1 in the reviewed period
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	none
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	2 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	none
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	2 in the reviewed period
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	1 in the reviewed period
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	none
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	1 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	1 in the reviewed period
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	1 in the reviewed period
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	none
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	none
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	2 in the reviewed period

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	2 in the reviewed period
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	2 in the reviewed period
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	1 in the reviewed period
Battery Management Service	charge scheduler	request-response interface	v4.11	not automated	1 in the reviewed period
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	2 in the reviewed period
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	none
Gripper Controller	safety controller fault input	published schema	v3.5	enforced on the producer side	2 in the reviewed period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	1 in the reviewed period
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	1 in the reviewed period
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	none
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	none
Event Bus	every consuming service	published schema	v1.5	advisory only	1 in the reviewed period
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	none
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	none
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	1 in the reviewed period
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	1 in the reviewed period
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	none
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	1 in the reviewed period
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	none
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	1 in the reviewed period
Alarm Router	paging provider	streamed topic	v2.3	not automated	none
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	2 in the reviewed period
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	none
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	1 in the reviewed period
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	2 in the reviewed period
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	none
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	none
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	none
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	1 in the reviewed period
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	2 in the reviewed period
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	1 in the reviewed period
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	none
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	none

3. Copyleft exposure and the shipped-software boundary

Chartkit reaches the customer only through Remote Operations Console, and the reviewer confirmed there is no second path. On provenance, chartkit shows public registry; upstream archived in 2025. No copyleft applies to chartkit, whose licence is BSD-3-Clause. chartkit is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The inventory entry OSS-010 is sqlite-embedded 3.45.1, on-robot local state. In the approval log, sqlite-embedded is approved 2023-06-15. The open question on sqlite-embedded is that no attribution obligation. sqlite-embedded was reviewed against the position twelve months earlier, and nothing measurable has changed.

Jwt-verify reaches the customer only through Identity and Authorization, and the reviewer confirmed there is no second path. In the approval log, jwt-verify is approved 2024-06-21. A build-time dependency whose absence would stop the pipeline rather than the product attaches to jwt-verify independently of its licence terms. jwt-verify was excluded from an earlier draft of TDR-2025-24 and reinstated after the reviewer obtained the supporting record.

No copyleft applies to csv-export, whose licence is MIT. Provenance of csv-export is public registry, signature verified. At OSS-026 the component is csv-export, pinned at 2.0.3 and used billing and usage exports. csv-export carries a follow-up request that was satisfied after the draft of TDR-2025-24 was circulated.

The licence on image-signing is Apache-2.0; copyleft exposure is no. OTA Updater is where image-signing is consumed, which determines whether distribution occurs at all. Version 2.5.0 of image-signing is the catalogued build at OSS-029, used artefact signature verification. image-signing illustrates why TDR-2025-24 separates the severity of a finding from the confidence behind it.

The non-licence risk on opencv-subset is a fork that must be maintained locally for as long as it is used. The approval position on opencv-subset is approved 2024-07-02. Provenance of opencv-subset is public registry, signature verified. opencv-subset is the clearest instance in TDR-2025-24 of engineering practice being better than the documentation suggests.

Zlib governs physics-core-fork, which the inventory marks as no copyleft. Physics-core-fork obliges the company to share alike any derived asset it distributes. Because physics-core-fork is used simulation physics, its licence terms reach the delivered software. physics-core-fork was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Version 1.9.4 of http-router is the catalogued build at OSS-022, used cloud service routing. The approval position on http-router is approved 2024-01-09. Http-router carries MIT terms, classified no for copyleft purposes. http-router illustrates why TDR-2025-24 separates the severity of a finding from the confidence behind it.

Tsdb-client presents no identified risk beyond ordinary version currency to the engineering team. Policy compliance for tsdb-client rests on its approval entry, which is approved 2024-09-11. Compliance with the tsdb-client terms requires to keep the dynamic-linking boundary intact. tsdb-client is not material on its own and belongs in this report because of what it reveals about the surrounding process.

In the approval log, warehouse-sim-assets is approved with conditions 2024-11-06. Warehouse-sim-assets obliges the company to hold a current commercial licence for each developer seat. Warehouse-sim-assets presents no identified risk beyond ordinary version currency to the engineering team. warehouse-sim-assets was checked against the corresponding entry in the company's own register, and the register was current.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2025-24)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	audit trail only
SEC-2025-02	High	142	30 days	no	restricted role membership
SEC-2025-03	Medium	87	90 days	yes	restricted role membership
SEC-2025-04	Medium	43	90 days	yes	network segmentation
SEC-2025-05	High	76	30 days	no	network segmentation

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-06	Medium	22	90 days	yes	manual review before release
SEC-2025-07	Medium	24	90 days	yes	manual review before release
SEC-2025-08	Medium	154	90 days	no	none recorded
SEC-2025-09	Medium	308	90 days	no	manual review before release
SEC-2025-10	Low	167	180 days	yes	audit trail only
SEC-2025-11	Low	78	180 days	yes	monitoring alert on use
SEC-2025-12	Low	354	180 days	no	restricted role membership
SEC-2026-01	High	232	30 days	no	restricted role membership
SEC-2026-02	High	127	30 days	no	monitoring alert on use
SEC-2026-03	High	27	30 days	yes	restricted role membership
SEC-2026-04	Medium	135	90 days	no	audit trail only
SEC-2026-05	Medium	100	90 days	no	audit trail only
SEC-2026-06	Medium	192	90 days	no	second delivery path
SEC-2026-07	Medium	158	90 days	no	monitoring alert on use
SEC-2026-08	Medium	151	90 days	no	none recorded
SEC-2026-09	Low	150	180 days	yes	none recorded
SEC-2026-10	Low	78	180 days	yes	restricted role membership
SEC-2026-11	Low	200	180 days	no	manual review before release
SEC-2026-12	Low	252	180 days	no	network segmentation
SEC-2026-13	Low	24	180 days	yes	network segmentation
SEC-2026-14	Low	128	180 days	yes	manual review before release
SEC-2026-15	Low	127	180 days	no	second delivery path
SEC-2026-16	Medium	148	90 days	no	none recorded
SEC-2026-17	High	147	30 days	no	second delivery path
SEC-2026-18	Low	305	180 days	no	second delivery path
SEC-2026-19	Medium	116	90 days	no	second delivery path
SEC-2026-20	Low	97	180 days	no	none recorded

4. Where third-party code sits inside the product

Anjali Sekhon of Applications maintains Remote Operations Console, and the service inventory entry was current at the review date. Remote Operations Console depends on the identity service, the telemetry ingest path and the video snippet service and is depended on by the traffic manager and the alarm router. Remote Operations Console pairs a TypeScript with a React front end implementation with cloud, a static bundle plus a session gateway, and its data lives in PostgreSQL 14 for session records and object storage for clips. Remote Operations Console carries 27 open defects, among them a case where the session list keeps rendering a robot that has already left the site until a manual refresh. Remote Operations Console was assessed against the objective and the defect record, and the component is adequate today and carries a supportability liability that is not on the funded plan. Remote Operations Console rests on a single source, and the reviewer has said so rather than implying corroboration.

Remote Operations Console publishes a stable interface, though two fields used by the traffic manager and the alarm router are documented only in code comments. Remote Operations Console releases twice a week. The evidence base for Remote Operations Console was the console interaction specification, session latency measurements and an accessibility walk-through. On end-of-life exposure, Remote Operations Console shows a charting library that no longer receives security updates. The manifests place Remote Operations Console cloud, with a thin session relay in each region. Remote Operations Console is the entry that changed most between the first and second document requests behind TDR-2025-24.

Reading the dependency map from Video Snippet Service outward gives the on-robot camera buffer and the identity service on the input side and the remote operations console and the incident review workflow on the output side. Video Snippet Service pairs a Go 1.22 with a media pipeline implementation with cloud, containers plus a media transcoder, and its data lives in object storage with a thirty-day lifecycle and a PostgreSQL 14 index. Interface discipline around Video Snippet Service is adequate; the schema is versioned and change is gated on a compatibility check. For Video Snippet Service the reviewer records that management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Where Video Snippet Service is concerned, the commitment is that a requested snippet is available to the operator inside 45 s; the reviewer traced the metric to its source. Video Snippet Service was tested rather than accepted, and the test is described in the working papers behind TDR-2025-24.

The evidence base for Video Snippet Service was the retention configuration, a lifecycle audit across regions and the access log sample. Automated test coverage on Video Snippet Service stands at 43 per cent line coverage. The open defect backlog for Video Snippet Service holds 12 items, the most consequential being that the retention lifecycle was applied per bucket, so one region kept clips for ninety days. The manifests place Video Snippet Service cloud, regional, with an on-robot capture buffer. Ownership of Video Snippet Service sits with Applications; Anjali Sekhon signs off changes to it. Video Snippet Service is reported at the evidence strength available and has not been promoted to fit the narrative.

The reviewer's judgement on Billing Usage Meter is that the design is sound and the test evidence is thinner than the design deserves. The ageing position for Billing Usage Meter is none identified, and the reviewer checked it against the vendor statements on file. The Data and Telemetry team carries Billing Usage Meter on its service inventory, and Nadia Semenova answered for it in interview. The topology for Billing Usage Meter is cloud, regional, and the reviewer confirmed it against the deployment manifests. Data and Telemetry releases Billing Usage Meter monthly, aligned to the billing cycle, and no release in the sampled window bypassed the gate set. Billing Usage Meter rests on a single source, and the reviewer has said so rather than implying corroboration.

Interface discipline around Billing Usage Meter is adequate; the schema is versioned and change is gated on a compatibility check. The finance export and the customer usage report rely on Billing Usage Meter, which in turn relies on the order broker and the WMS connector. Billing Usage Meter sits at 79 per cent line coverage; the trend across the reviewed period was up by seven points. Billing Usage Meter carries 7 open defects, among them a case where a re-run of a failed metering window can double-count if the operator skips the reset step. Implementation of Billing Usage Meter rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to PostgreSQL 14 with an immutable usage ledger. Billing Usage Meter was one of two items where TDR-2025-24 could not obtain the underlying data within the engagement.

Deployment of Lidar Driver is edge, one process per sensor, which sets the blast radius of any defect in it. For Lidar Driver the supportability position is the vendor's software development kit is two major versions behind current. Robot Firmware releases Lidar Driver as needed, historically twice a year, and no release in the sampled window bypassed the gate set. Lidar Driver is owned by the Robot Firmware team, with Yuki Harada named as the engineering owner of record. Coverage for Lidar Driver was 44 per cent line coverage at the review date, and the trend over the reviewed period was down by two points after a large refactor. Lidar Driver is the entry that changed most between the first and second document requests behind TDR-2025-24.

18 defects remain open against Lidar Driver; the one the reviewer pressed on is that reconnect handling leaves the timestamp offset uncorrected for the first three frames. Behind Lidar Driver sits no store; frames are handed to fusion in memory; the service itself is C++17 on on-robot Linux, user-space driver. Where Lidar Driver is concerned, the commitment is that no more than one dropped scan per ten thousand at nominal spin rate; the reviewer traced the metric to its source. The interface Lidar Driver publishes is versioned, and the costmap fusion pipeline and the localization engine were checked for use of any undocumented field. What the reviewer had for Lidar Driver was the driver source, the vendor integration guide and a captured reconnect trace; the conclusions below do not reach past it. Lidar Driver is the clearest instance in TDR-2025-24 of engineering practice being better than the documentation suggests.

The Developer Productivity team carries Release Orchestrator on its service inventory, and Colin Achebe answered for it in interview. Automated test coverage on Release Orchestrator stands at 55 per cent line coverage. 16 defects remain

open against Release Orchestrator; the one the reviewer pressed on is that an emergency release path allows a single approver, and it was used more often than the policy expects. Release Orchestrator depends on the build system, the simulation harness gate and the identity service and is depended on by the OTA updater and the cloud deployment controllers. Release Orchestrator was weighed on its coverage, its defect backlog and its dependency position, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Release Orchestrator is included because a reader of TDR-2025-24 should be able to disagree with the inference without repeating the work.

Hosting for Release Orchestrator is cloud, a pipeline runner with approval hooks, and its Go 1.22 code writes through to PostgreSQL 14 for release records and object storage for artefacts. End-of-life review of Release Orchestrator found none identified. Data ownership is clear for Release Orchestrator: it writes PostgreSQL 14 for release records and object storage for artefacts and no other service writes there. The objective recorded against Release Orchestrator is that no artefact reaches production without a recorded approval and a passing gate set, and it is reported weekly to the engineering leadership. What the reviewer had for Release Orchestrator was the release policy, ninety release records and an audit of emergency-path usage; the conclusions below do not reach past it. Release Orchestrator is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The manifests place Charge Scheduler cloud, regional, with a conservative local fallback policy on the robot. The backlog for Charge Scheduler is 7 items deep and contains an acknowledged defect whereby opportunity-charge windows are not released when a shift ends earlier than planned. Charge Scheduler has a single named owner, Nadia Semenova of Fleet Services, which is not true of every service reviewed. Charge Scheduler releases fortnightly, without a formal approval gate until March 2026. The dependency map places the battery management service and the site shift calendar above Charge Scheduler and the fleet allocator and the fleet health monitor below it. Charge Scheduler is treated as material to the investment case rather than as a routine engineering item.

The published objective for Charge Scheduler is that fleet state of charge stays above 35 per cent through the peak shift at every site. On Charge Scheduler the conclusion is qualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Automated test coverage on Charge Scheduler stands at 63 per cent line coverage. Charge Scheduler was examined for unsupported components; none; the dependency set was refreshed in the 2026 platform upgrade. Charge Scheduler publishes a stable interface, though two fields used by the fleet allocator and the fleet health monitor are documented only in code comments. Charge Scheduler has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Documents and data inspected for Edge Gateway Agent were the agent design note, buffer sizing calculations and a four-hour offline test. The Platform Infrastructure team carries Edge Gateway Agent on its service inventory, and Sam Oyelaran answered for it in interview. Interface discipline around Edge Gateway Agent is adequate; the schema is versioned and change is gated on a compatibility check. Edge Gateway Agent is promoted every three weeks, with the ring order fixed by policy. Edge Gateway Agent sits edge, one per robot, with mutual authentication to the cloud, and no alternative placement is currently supported. Edge Gateway Agent is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Edge Gateway Agent shows 74 per cent line coverage in the pipeline, and the figure is respectable but excludes the fallback path, which is where the known defect sits. Edge Gateway Agent carries 5 open defects, among them a case where buffer pressure is reported as a robot-level metric and is not aggregated to a site view. The dependency map places the on-robot services that emit telemetry above Edge Gateway Agent and the telemetry ingest path and the robot shadow store below it. Edge Gateway Agent is written in Rust 1.79, executes on on-robot Linux, a single supervised process, and keeps its state in a local write-ahead buffer sized for four hours. On Edge Gateway Agent the conclusion is qualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Edge Gateway Agent was compared with the equivalent position at two other components, and it is neither the best nor the worst.

11 items sit in the Order Broker backlog; the entry that bears on the investment case is that order cancellation races with assignment and can leave an orphaned task record. For Order Broker the reviewer records that it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Order Broker does not share a schema with any consumer; every read by the fleet allocator and the analytics warehouse loader goes through the published interface. Order Broker's dependency ageing was checked: none identified. Upstream of Order Broker sit the

WMS connector and the site configuration service; downstream, the fleet allocator and the analytics warehouse loader consume its output. Order Broker would move from a concern to a verified defect if one further measurement were supplied.

Documents and data inspected for Order Broker were the broker journal schema, the nightly reconciliation output and two cancellation incidents. The published objective for Order Broker is that no accepted order is lost, measured by journal reconciliation each night. The manifests place Order Broker cloud, regional. Order Broker has a single named owner, Grace Lindqvist of Fleet Services, which is not true of every service reviewed. Order Broker was built in Go 1.22 and has not been rewritten; it still runs on cloud, containers on a managed Kubernetes cluster against PostgreSQL 14 with an event journal. Order Broker was excluded from an earlier draft of TDR-2025-24 and reinstated after the reviewer obtained the supporting record.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-24)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-410 manipulator	network radio	1.17.8	9	rig one only	pre-production
HR-410 manipulator	wheel encoder set	1.6.4	11	neither rig	current
HR-200 tote mover	inertial measurement unit	2.9.0	10	both rigs	pre-production
HR-240 tote mover	safety scanner	4.10.5	6	rig one only	one revision behind
HR-320 pallet mover	navigation lidar	1.15.8	14	neither rig	superseded
HR-300 pallet mover	wheel encoder set	4.3.2	2	both rigs	current
HR-240 tote mover	depth camera	3.10.8	4	neither rig	one revision behind
HR-410 manipulator	safety scanner	3.5.6	12	neither rig	current
HR-240 tote mover	load cell deck	4.4.3	10	rig one only	pre-production
HR-200 tote mover	wheel encoder set	1.6.7	8	neither rig	superseded
HR-240 tote mover	safety scanner	2.14.1	12	neither rig	one revision behind
HR-300 pallet mover	navigation lidar	4.11.5	2	rig two only	current
HR-320 pallet mover	navigation lidar	4.10.2	9	neither rig	current
HR-240 tote mover	inertial measurement unit	1.12.1	13	both rigs	superseded
HR-300 pallet mover	wheel encoder set	2.5.5	14	neither rig	pre-production
HR-240 tote mover	inertial measurement unit	4.1.7	11	rig two only	current
HR-320 pallet mover	safety scanner	4.2.7	4	rig two only	current
HR-410 manipulator	gripper coprocessor	1.9.7	6	rig one only	pre-production
HR-240 tote mover	safety scanner	1.9.1	6	rig two only	current
HR-410 manipulator	depth camera	2.17.6	8	rig two only	current
HR-320 pallet mover	safety scanner	2.9.7	5	neither rig	one revision behind
HR-200 tote mover	network radio	3.1.0	9	rig one only	one revision behind

4. Where third-party code sits inside the product - part 2 of 2

Where the 4. Where third-party code sits inside the product was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. The break is editorial.

Treat the conventions declared for the 4. Where third-party code sits inside the product as governing here too; this part introduces none of its own. Pagination alone determines where it falls.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-24) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-410 manipulator	load cell deck	1.2.1	14	both rigs	superseded
HR-300 pallet mover	wheel encoder set	3.10.1	11	rig one only	pre-production
HR-240 tote mover	wheel encoder set	1.18.2	10	rig two only	one revision behind
HR-300 pallet mover	navigation lidar	4.19.1	7	both rigs	pre-production
HR-410 manipulator	E-stop harness	4.9.0	2	neither rig	pre-production
HR-320 pallet mover	compute module	2.16.1	4	rig one only	pre-production
HR-200 tote mover	inertial measurement unit	4.3.3	8	rig two only	pre-production
HR-320 pallet mover	charge contact assembly	4.6.0	8	rig two only	pre-production
HR-200 tote mover	safety scanner	2.6.0	13	rig two only	pre-production
HR-200 tote mover	depth camera	2.18.2	11	rig two only	superseded
HR-300 pallet mover	inertial measurement unit	2.18.2	8	rig one only	superseded
HR-300 pallet mover	depth camera	4.6.8	2	rig two only	pre-production

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-240 tote mover	compute module	3.14.2	3	rig two only	pre-production
HR-240 tote mover	inertial measurement unit	1.5.1	14	rig one only	pre-production
HR-300 pallet mover	depth camera	3.4.5	5	rig one only	one revision behind
HR-200 tote mover	gripper coprocessor	2.9.2	2	rig two only	current
HR-240 tote mover	inertial measurement unit	3.6.2	2	both rigs	current
HR-320 pallet mover	E-stop harness	4.17.3	14	rig one only	current
HR-300 pallet mover	safety scanner	4.16.2	6	rig one only	superseded
HR-240 tote mover	charge contact assembly	3.11.4	2	both rigs	one revision behind
HR-300 pallet mover	depth camera	2.18.2	7	rig two only	one revision behind
HR-410 manipulator	navigation lidar	4.4.7	13	rig one only	pre-production

5. Provenance, approval and contributor assignment

Nav-graph-solver reaches the customer only through Motion Planner, and the reviewer confirmed there is no second path. Nav-graph-solver presents a fork that must be maintained locally for as long as it is used to the engineering team. Public registry, signature verified is the provenance recorded for nav-graph-solver. nav-graph-solver was one of two items where TDR-2025-24 could not obtain the underlying data within the engagement.

Media-transcode version 6.0.0 is catalogued at OSS-016 and is used video snippet transcoding. Weak copyleft applies to media-transcode, whose licence is LGPL-2.1 with an optional GPL build. The non-licence risk on media-transcode is a component with open advisories that do not affect the deployed configuration. media-transcode was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The approval position on grid-render is approved with conditions 2023-07-19. In addition to its terms, grid-render brings a single-maintainer dependency with no succession plan. Grid-render version 3.3.5 is catalogued at OSS-021 and is used map authoring geometry rendering. grid-render was excluded from an earlier draft of TDR-2025-24 and reinstated after the reviewer obtained the supporting record.

Licence text and the actual usage of lidar-vendor-sdk were read together, and two major versions behind current; the vendor's support statement covers the deployed version to 2027. The distribution question for lidar-vendor-sdk turns on the fact that it is used lidar driver. The approval position on lidar-vendor-sdk is approved 2023-05-22. lidar-vendor-sdk was tested rather than accepted, and the test is described in the working papers behind TDR-2025-24.

Because yaml-parse is used configuration parsing on the robot, its licence terms reach the delivered software. Usage of yaml-parse 0.15.2 is configuration parsing on the robot, and the inventory reference is OSS-013. Yaml-parse carries MIT terms, classified no for copyleft purposes. yaml-parse was excluded from an earlier draft of TDR-2025-24 and reinstated after the reviewer obtained the supporting record.

Beyond licensing, k8s-operator-framework carries a fork that must be maintained locally for as long as it is used. K8s-operator-framework reaches the customer only through Site Provisioning Service, and the reviewer confirmed there is no second path. OSS-028 records k8s-operator-framework 1.31.0 against Site Provisioning Service, cloud deployment controllers. k8s-operator-framework was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Approved 2024-05-14 is the approval entry for otel-instrument. On otel-instrument the reviewer records that attribution present. Otel-instrument entered the codebase by this route: public registry, signature verified. otel-instrument is the clearest instance in TDR-2025-24 of engineering practice being better than the documentation suggests.

Linking mode matters for wms-adapter-sdk, which is used one customer's warehouse system adapter. Where wms-adapter-sdk came from matters here: supplied by the warehouse system vendor under a written licence. OSS-017 records wms-adapter-sdk 2.2.0 against WMS Connector, one customer's warehouse system adapter. wms-adapter-sdk is reported here in full because a summary of it in an earlier draft of TDR-2025-24 was misread as an assurance.

Geom2d-fusion is licensed under LGPL-2.1, with copyleft exposure recorded as weak. The approval position on geom2d-fusion is approved 2023-08-04. In Costmap Fusion the component geom2d-fusion is used dynamically linked by the fusion pipeline, which is the fact the licence analysis rests on. geom2d-fusion rests on a single source, and the reviewer has said so rather than implying corroboration.

The open question on rtos-shim is that no obligations beyond attribution. Rtos-shim was assessed for support as well as licence risk, and it carries an upstream project that is active but changes interface frequently. Rtos-shim appears at OSS-008 in version 1.0.9; the usage recorded is real-time helper layer on the robot. rtos-shim was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2025-24)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-07-01	the objective is met on the company's own measurement only	moderate	narrows the conclusion to the tested scale	a re-run of the original test
OQ-07-02	the measurement basis changed during the reviewed period	limited	does not change the conclusion	one further interview
OQ-07-03	the second bench rig does not cover this path	strong	narrows the conclusion to the tested scale	one further interview
OQ-07-04	the supplier certificate was read rather than the test witnessed	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-07-05	the remediation landed without a regression test	moderate	is reflected in the post-close plan	a document the company has not produced
OQ-07-06	the declined item has no expiry date	strong	would change the conclusion if resolved adversely	a re-run of the original test
OQ-07-07	two teams each believe the other owns this	strong	does not change the conclusion	one further interview
OQ-07-08	two teams each believe the other owns this	limited	is reflected in the post-close plan	an independent measurement
OQ-07-09	the objective is met on the company's own measurement only	moderate	does not change the conclusion	a re-run of the original test
OQ-07-10	the measurement basis changed during the reviewed period	moderate	narrows the conclusion to the tested scale	a decision from management
OQ-07-11	the compensating control was not tested	moderate	does not change the conclusion	a re-run of the original test
OQ-07-12	the remediation landed without a regression test	limited	narrows the conclusion to the tested scale	one further interview
OQ-07-13	the second bench rig does not cover this path	limited	is reflected in the post-close plan	one further interview
OQ-07-14	the retention window closed before the raw data was requested	strong	would change the conclusion if resolved adversely	a re-run of the original test
OQ-07-15	the compensating control was not tested	limited	would change the conclusion if resolved adversely	a re-run of the original test
OQ-07-16	the compensating control was not tested	moderate	does not change the conclusion	one further interview
OQ-07-17	the supplier certificate was read rather than the test witnessed	limited	leaves one section unconcluded	a document the company has not produced
OQ-07-18	the measurement basis changed during the reviewed period	strong	narrows the conclusion to the tested scale	a witnessed test
OQ-07-19	the remediation landed without a regression test	strong	would change the conclusion if resolved adversely	a witnessed test
OQ-07-20	the second bench rig does not cover this path	moderate	does not change the conclusion	a witnessed test
OQ-07-21	the second bench rig does not cover this path	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-07-22	the remediation landed without a regression test	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-07-23	the population completeness could not be proven	moderate	would change the conclusion if resolved adversely	a decision from management
OQ-07-24	the remediation landed without a regression test	strong	leaves one section unconcluded	one further interview
OQ-07-25	the retention window	strong	does not change the	a document the company

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
	closed before the raw data was requested		conclusion	has not produced
OQ-07-26	the population completeness could not be proven	limited	would change the conclusion if resolved adversely	a document the company has not produced
OQ-07-27	the objective is met on the company's own measurement only	moderate	would change the conclusion if resolved adversely	a document the company has not produced
OQ-07-28	the population completeness could not be proven	limited	would change the conclusion if resolved adversely	a document the company has not produced
OQ-07-29	the retention window closed before the raw data was requested	limited	would change the conclusion if resolved adversely	a decision from management
OQ-07-30	the measurement basis changed during the reviewed period	moderate	does not change the conclusion	a decision from management
OQ-07-31	the retention window closed before the raw data was requested	strong	leaves one section unconcluded	a document the company has not produced
OQ-07-32	the compensating control was not tested	strong	leaves one section unconcluded	a decision from management

6. Findings arising from dependency and secret scanning

A calibration change cannot be reconstructed during an investigation is the exposure finding SEC-2026-20 describes. For Low-severity finding SEC-2026-20 the record shows that record edited on a laboratory robot, which is strong evidence and is graded separately from severity. What remains after SEC-2026-20 is remediated is low; management accepted pending the calibration service redesign. Finding SEC-2026-20 was opened on 2026-04-24 and states that payload scale calibration record can be edited without an audit entry. assessment finding SEC-2026-20 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Assessment finding SEC-2025-06 was graded Medium, band 6.5, by Redwater Security engagement RW-2025-3. The reviewer declines to soften finding SEC-2025-06: the finding was closed within the committed window and the closure evidence was inspected. SEC-2025-06 against Video Snippet Service reduces to a residual risk of low; the index is scoped per tenant and the identifiers are opaque. On 2025-09-19 the tester recorded assessment finding SEC-2025-06: video snippet index allowed cross-tenant enumeration by identifier. assessment finding SEC-2025-06 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

For finding SEC-2026-05 the record shows that credential reuse confirmed from the secret store inventory, which is moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence and is graded separately from severity. On finding SEC-2026-05 the conclusion is that the fix is verified and the class of defect is not, since no test now prevents its return. Open is the status of finding SEC-2026-05, dated committed 2026-09-30. SEC-2026-05 is rated Medium with a numeric band of 5.2 on the tester's scoring scale. Alarm Router finding SEC-2026-05 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Low-severity finding SEC-2025-11 sits against Remote Operations Console and records that console session cookie lacked a strict same-site attribute. On finding SEC-2025-11 the tester's evidence was that attribute absence confirmed; no working exploit chain was demonstrated, and the reviewer grades it strong. Assessment finding SEC-2025-11 produces this judgement: the finding was closed within the committed window and the closure evidence was inspected. Rating for assessment finding SEC-2025-11 is Low, band 3.1; the reviewer re-read the scoring rationale and accepted it. Remote Operations Console finding SEC-2025-11 was reviewed against the position twelve months earlier, and nothing measurable has changed.

The reviewer's position on assessment finding SEC-2026-04 is that the finding was closed within the committed window and the closure evidence was inspected. Finding SEC-2026-04 sits against WMS Connector and records that wMS connector logged customer order payloads at debug level. SEC-2026-04 against WMS Connector matters because customer commercial data is retained in operational logs. As at the review date SEC-2026-04 against WMS Connector

was remediated and the closure entry reads 2026-05-20. WMS Connector finding SEC-2026-04 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The practical consequence in Edge Gateway Agent finding SEC-2026-12 is that a disposed compute module carries a usable fleet credential. Edge Gateway Agent finding SEC-2026-12 stands partially remediated; pending is what the record gives for closure. The residual position on SEC-2026-12 against Edge Gateway Agent is medium; a wipe procedure exists but was not applied retroactively, stated after mitigation rather than assuming full success. Finding SEC-2026-12 was opened on 2026-04-20 and states that edge gateway agent retained mutual authentication keys after decommissioning. Edge Gateway Agent finding SEC-2026-12 is treated as material to the investment case rather than as a routine engineering item.

Finding SEC-2025-10 matters because site naming and fleet size leak to anyone inside the cluster network. SEC-2025-10 against Charge Scheduler leaves residual risk that the reviewer assesses as low. Moderate evidence supports Charge Scheduler finding SEC-2025-10, namely that endpoint reachable and content captured. At 3.7 the score on assessment finding SEC-2025-10 places it in the Low group. SEC-2025-10 against Charge Scheduler was inspected in the repository rather than described from the ticket, which is why the account is specific.

SEC-2026-02 against Robot Shadow Store is not fully retired by its remediation: low; authorization is checked on both the cache and durable paths. Finding SEC-2026-02 was opened on 2026-04-15 and states that robot shadow store cache lacked authorization on the internal read path. On SEC-2026-02 against Robot Shadow Store the tester's evidence was that the tester read another tenant's shadow state from a test workload, and the reviewer grades it moderate, because the demonstration was performed in a laboratory tenant rather than in production. Severity on SEC-2026-02 against Robot Shadow Store is High, scored 7.6 by the assessor. finding SEC-2026-02 was raised again with the owning engineer after the initial interview, and the position did not change.

For SEC-2026-06 against Simulation Harness the record shows that role policy read from the cloud configuration export, which is moderate evidence and is graded separately from severity. The residual position on assessment finding SEC-2026-06 is medium; the role was scoped down but still reads one production bucket, stated after mitigation rather than assuming full success. Severity on Medium-severity finding SEC-2026-06 is Medium, scored 6.1 by the assessor. Simulation harness cluster used a long-lived cloud role with broad read access is the substance of SEC-2026-06 against Simulation Harness, raised on 2026-04-17. SEC-2026-06 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Fleet Health Monitor finding SEC-2026-09 stands remediated; 2026-05-08 is what the record gives for closure. The residual position on Fleet Health Monitor finding SEC-2026-09 is low, stated after mitigation rather than assuming full success. The practical consequence in assessment finding SEC-2026-09 is that a single query can exhaust the service's memory budget. At 3.9 the score on finding SEC-2026-09 places it in the Low group. SEC-2026-09 was one of two items where TDR-2025-24 could not obtain the underlying data within the engagement.

In SEC-2025-01 the exploitation path means a captured token remains usable long after the session that produced it. Finding SEC-2025-01 leaves residual risk that the reviewer assesses as low once lifetimes are cut to one hour with refresh. For SEC-2025-01 the record shows that reproduced by the tester and confirmed in the token audit, which is moderate evidence and is graded separately from severity. Finding SEC-2025-01 stands remediated; 2025-11-14 is what the record gives for closure. finding SEC-2025-01 has no single accountable owner, and that is the finding the reviewer would press hardest.

What remains after Billing Usage Meter finding SEC-2026-10 is remediated is low; object lock is applied and verified nightly. The reason SEC-2026-10 against Billing Usage Meter was not downgraded is that an export could be altered after the fact. The evidence for SEC-2026-10 is limited, resting on a configuration read rather than on a demonstrated exploitation: bucket configuration read from the cloud export. Low-severity finding SEC-2026-10 is rated Low with a numeric band of 4.0 on the tester's scoring scale. assessment finding SEC-2026-10 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Read alongside the remediation record, Low-severity finding SEC-2025-12 shows that the item is open past its committed date and no revised date has been agreed. Low is the rating Analytics Warehouse Loader finding SEC-2025-12 carries, at 4.2 on the assessor's own scale. Analytics Warehouse Loader finding SEC-2025-12 is not fully retired by its

remediation: medium; management accepted the risk with a compensating audit trail. The evidence for assessment finding SEC-2025-12 is moderate: permissions read from the warehouse grant inventory. SEC-2025-12 was excluded from an earlier draft of TDR-2025-24 and reinstated after the reviewer obtained the supporting record.

The remediation ledger shows Map Authoring Studio finding SEC-2025-04 as remediated, 2025-11-27. On finding SEC-2025-04 the tester's evidence was that reproduced, and independently observed in a field incident, and the reviewer grades it strong. Finding SEC-2025-04 produces this judgement: the item is open past its committed date and no revised date has been agreed. Medium is the rating Medium-severity finding SEC-2025-04 carries, at 5.4 on the assessor's own scale. SEC-2025-04 was checked against the corresponding entry in the company's own register, and the register was current.

7. Remediation of the licence exposure and its cost

About register item TD-07 it should be said that the check is written but not enabled as at the review date. Tooling item TD-07 is on the register because it non-reproducible test results undermine every simulation claim. For Simulation Harness item TD-07 the team estimates 0.5 engineer-quarters, which the reviewer converted to $68,000 for comparison. programme item TD-07 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

TD-09 on Event Bus is the item to upgrade the streaming platform before its support window closes, classified under support. The consequence of leaving register item TD-09 undone is that it an unsupported broker underneath every service. Effort on programme item TD-09 is estimated at 1.5 engineer-quarters, costed at $196,000. Event Bus item TD-09 was cross-checked against the engineering record supplied for TDR-2025-24, and the two agree.

TD-13 on Localization Engine cannot start earlier than it does because it is behind a customer maintenance window that is not yet agreed. Funding for performance item TD-13 is funded and already staffed, and the register does not say so. Left in place, the condition behind performance item TD-13 relocalisation exceeds the operator prompt timeout at sparse sites. TD-13 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Hardware-in-the-Loop Rig item TD-05 cannot start earlier than it does because it is behind a platform change that has its own target date. Hardware-in-the-Loop Rig item TD-05 is committed with a target date of 2026-09-30. Test capability item TD-05 is sized at 0.5 engineer-quarters, or $96,000 at the company's loaded rate. TD-05 on Hardware-in-the-Loop Rig was reviewed against the position twelve months earlier, and nothing measurable has changed.

The plan treats register item TD-04 as not funded, and carried in the register regardless, which the reviewer checked against the headcount model. For quality item TD-04 the team estimates 2 engineer-quarters, which the reviewer converted to $240,000 for comparison. In progress is the status of quality item TD-04, targeted at 2027-03-31. TD-04 is one of the entries where TDR-2025-24 relies on the company's own measurement rather than an independent one.

Charge Scheduler item TD-23 is funded from the second quarter onward; the register status and the funding position are not the same thing. The register shows TD-23 on Charge Scheduler as committed, due 2026-12-31. Sequencing of register item TD-23 matters: it is behind a supplier release outside the company's control. reliability item TD-23 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-24.

Committed is the status of Analytics Warehouse Loader item TD-10, targeted at 2026-12-31. Item programme item TD-10 in the debt register is to migrate the deprecated warehouse transformation function, and it bears on Analytics Warehouse Loader. The warehouse vendor has published a withdrawal date is the point the reviewer would put to management on register item TD-10. TD-10 on Analytics Warehouse Loader is reported here in full because a summary of it in an earlier draft of TDR-2025-24 was misread as an assurance.

Item register item TD-21 in the debt register is to refit the battery state-of-health estimator on opportunity-charge data, and it bears on Battery Management Service. Programme item TD-21 is behind a platform change that has its own target date, which is why the target date is later than the effort estimate implies. Progress on TD-21 is recorded as in progress against a 2026-09-30 date. programme item TD-21 is reported here in full because a summary of it in an earlier draft of TDR-2025-24 was misread as an assurance.

The reviewer's qualification on TD-19 on Speed and Separation Monitor is that the most significant open safety item in the reviewed record. Behind a platform change that has its own target date is where Speed and Separation Monitor

item TD-19 sits in the programme. Programme item TD-19 is unfunded and dependent on capital rather than engineering effort; the register status and the funding position are not the same thing. programme item TD-19 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Left in place, the condition behind TD-20 drift on wet floors is neither corrected nor flagged. Funding for programme item TD-20 is funded in the current financial year, and the register does not say so. TD-20 covers the work to validate and ship the wheel slip detector. Wheel Odometry Filter item TD-20 is one of the entries where TDR-2025-24 relies on the company's own measurement rather than an independent one.

The consequence of leaving TD-06 on Simulation Harness undone is that it a nightly regression can reach a release candidate. Effort on register item TD-06 is estimated at 1 engineer-quarter, costed at $115,000. About register item TD-06 it should be said that blocked on the physics core determinism work. programme item TD-06 is reported here in full because a summary of it in an earlier draft of TDR-2025-24 was misread as an assurance.

The reviewer's comment on register item TD-18 is that the safety team supports it; it is not funded in the current plan. Backlog is the status of programme item TD-18, targeted at 2027-03-31. Register item TD-18 is sized at 0.75 engineer-quarters, or $104,000 at the company's loaded rate. programme item TD-18 was raised again with the owning engineer after the initial interview, and the position did not change.

Status on programme item TD-08 is backlog; the target is 2027-01-31. Register item TD-08 is sized at 1 engineer-quarter, or $132,000 at the company's loaded rate. Whether programme item TD-08 happens depends on funding, and it is funded from the second quarter onward. security supportability item TD-08 is included because a reader of TDR-2025-24 should be able to disagree with the inference without repeating the work.

The dependency structure places TD-11 on Site Provisioning Service behind a platform change that has its own target date. Sizing on register item TD-11 gives 0.5 engineer-quarters and $61,000. Item TD-11 in the debt register is to re-converge the four forked site provisioning templates, and it bears on Site Provisioning Service. Site Provisioning Service item TD-11 carries a follow-up request that was satisfied after the draft of TDR-2025-24 was circulated.

8. Incidents involving a third-party component

Contributing to configuration drift incident INC-2026-051, and not repaired by the remediation, is that template drift was not detected because the baseline check compares only the default template. Larkspur Home incident INC-2026-051 traces back to a single fact: the site template had been forked for a customer network requirement and drifted from the baseline. Reported detection applies to configuration drift event INC-2026-051, since the identity service rejected the site's first token request. Following incident INC-2026-051 the team undertook to check every template against the baseline and re-converge the four forked templates, with Marcus Duplessis accountable by 2026-06-15. Duration for configuration drift event INC-2026-051 splits into 2 h 18 m of impact and 9 days to close the underlying cause. configuration drift event INC-2026-051 is the reason one section of TDR-2025-24 concludes narrowly rather than generally.

In taxonomy terms configuration drift incident INC-2026-051 is a configuration drift incident in Site Provisioning Service. Assessed on its own record, Site Provisioning Service incident INC-2026-051 shows that management's severity rating is defensible and its remediation priority is not. Site Provisioning Service incident INC-2026-051 affected 1 site or tenant with no robot taken out of service. The trigger recorded for INC-2026-051 on 2026-05-06 was that a new site went live with a bespoke network template that omitted the outbound certificate pin. INC-2026-051 closure status is the check landed on 2026-05-15; two templates remain forked. configuration drift incident INC-2026-051 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

On delivery, INC-2026-045 shows this: the alert landed on 2026-05-17; the buffer extension is in progress. 2026-04-21 is the date of observability incident INC-2026-045, on which a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. INC-2026-045 took 5 h 12 m to stabilise and 26 days to close out. Beyond the direct cause, Evergreen Warehousing incident INC-2026-045 shows that buffer pressure was never aggregated to a site view, so the condition was not observable. Sam Oyelaran committed after incident INC-2026-045 to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current hardware, due 2026-06-30. incident INC-2026-045 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Attention came to observability event INC-2026-045 when the buffer pressure metric was visible per robot but no site-level alert existed. For observability event INC-2026-045 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. The affected population in Evergreen Warehousing incident INC-2026-045 was one site and 39 robots. What observability incident INC-2026-045 tells the reviewer is that the team's analysis was honest about a gap it could have concealed. The classification applied to observability incident INC-2026-045 is observability, and the affected component is Edge Gateway Agent. observability event INC-2026-045 is the entry that changed most between the first and second document requests behind TDR-2025-24.

On delivery, incident INC-2026-021 shows this: not landed; the determinism check is written but not enabled. INC-2026-021 reached the engineering team when the autonomy team noticed a step change in the pass rate. Limited; the log retention window had closed before the reviewer asked for the raw data is the evidence grade for incident INC-2026-021, and the reader should not read severity into it. The engineering team's root cause for internal incident INC-2026-021 is that the physics core is not deterministic across optimisation levels, and the reviewer found the analysis sound. Drawing the evidence in tooling incident INC-2026-021 together, the blast radius was set by a deployment topology decision made years earlier. tooling incident INC-2026-021 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The immediate event in INC-2026-021 was that a nightly simulation suite produced non-reproducible results after the physics core was rebuilt, dated 2026-02-24. Not applicable; developer tooling elapsed before tooling event INC-2026-021 was mitigated, with resolution at open at the review date. In taxonomy terms Simulation Harness incident INC-2026-021 is a tooling incident in Simulation Harness. Internal incident INC-2026-021 affected no production site and no robot directly, the exposure being to internal systems. The commitment register shows Simulation Harness incident INC-2026-021 against an undertaking to pin the full build configuration and add a determinism check to the harness, with Colin Achebe named and 2026-05-31 committed. tooling event INC-2026-021 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

In incident INC-2026-009, the service workflow allowed the step to be marked complete without a calibration record, which is a process finding rather than a code finding. What Gripper Controller incident INC-2026-009 tells the reviewer is that the customer-reported detection path is the finding, not the defect itself. The trigger recorded for calibration incident INC-2026-009 on 2026-01-27 was that a gripper applied more force than the configured limit after a mechanical replacement without recalibration. 1 h 30 m to quarantine the affected robots elapsed before calibration incident INC-2026-009 was mitigated, with resolution at 48 days. Following Blue Mesa Components incident INC-2026-009 the team undertook to require a calibration record before returning a manipulator robot to service, with Ines Duarte accountable by 2026-04-15. calibration event INC-2026-009 is included because a reader of TDR-2025-24 should be able to disagree with the inference without repeating the work.

Strong, resting on the production traces the team supplied is the evidence grade for calibration event INC-2026-009, and the reader should not read severity into it. Blue Mesa Components incident INC-2026-009 was contained to one site and 4 robots. Gripper Controller owns incident INC-2026-009, which the reviewer classes as calibration. Verification of the Blue Mesa Components incident INC-2026-009 remediation gives landed on 2026-03-16 with a workflow gate. Incident INC-2026-009 was detected because the payload damage was reported by the customer. calibration event INC-2026-009 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

INC-2025-047 rests on limited; the log retention window had closed before the reviewer asked for the raw data evidence, which the reviewer states separately from the severity of the finding. In Lidar Driver incident INC-2025-047 the first signal was an automated fusion timing alert fired inside four minutes, and the reviewer treats that as alerted detection. The driver fix landed on 2025-04-05; the second rig was still unmodified at review is the position on the timing event INC-2025-047 commitment as at the review date. Asterion Logistics incident INC-2025-047 is not read as an isolated failure: the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. INC-2025-047 classifies as a timing event against Lidar Driver. INC-2025-047 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Incident INC-2025-047 began on 2025-04-02 when a lidar unit reconnected after a cable seating fault and reported frames with a stale timestamp offset. The corrective action for INC-2025-047 is to re-derive the time offset on every reconnect and add the affected variant to the second bench rig, and 2025-06-15 is the date the team gave. Timing event INC-2025-047 would have been shorter or smaller had it not been that the bench rig that would have caught it carries a different lidar variant. Root cause in timing incident INC-2025-047 was that the driver did not re-derive the sensor time offset on reconnect. 38 m elapsed before Asterion Logistics incident INC-2025-047 was mitigated, with resolution at 3 days. timing event INC-2025-047 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Verification of the process control event INC-2026-062 remediation gives not landed; the change is scoped but not scheduled. Process control event INC-2026-062 took not applicable; process only to stabilise and open at the review date to close out. For INC-2026-062 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. Drawing the evidence in process control event INC-2026-062 together, the failure was in the detection design rather than in the code that failed. In process control incident INC-2026-062, the policy defines an emergency but the tool does not ask which condition applies, which is a process finding rather than a code finding. Release Orchestrator incident INC-2026-062 was checked against the company's own register, which had not been updated since the change landed.

The detection path for incident INC-2026-062 was the release audit found it during the quarterly review. Process control incident INC-2026-062 touched no production site and no robot directly, the exposure being to internal systems, and internal was the affected account. Following Release Orchestrator incident INC-2026-062 the team undertook to require a justification and a second approver within one working day for every emergency release, with Colin Achebe accountable by 2026-08-31. Internal incident INC-2026-062 was caused by a design gap: the emergency path is available to any release engineer without a justification field. The classification applied to INC-2026-062 is process control, and the affected component is Release Orchestrator. internal incident INC-2026-062 is the reason one section of TDR-2025-24 concludes narrowly rather than generally.

Cause analysis in incident INC-2025-084 concluded that the pause flag was held in process memory rather than in the release state record. The trigger recorded for state persistence event INC-2025-084 on 2025-07-08 was that a paused update ring resumed after a routine service restart and eleven robots took an unintended firmware step. The detection path for state persistence event INC-2025-084 was the site team escalated when robots rebooted outside the maintenance window. OTA Updater incident INC-2025-084 produced one commitment: to persist the pause state and require a design note for any change to the release state machine, owned by Erik Palmgren, dated 2025-08-15. Evidence strength on the incident INC-2025-084 analysis is limited; the log retention window had closed before the reviewer asked for the raw data. INC-2025-084 carries a follow-up request that was satisfied after the draft of TDR-2025-24 was circulated.

OTA Updater accounts for incident INC-2025-084, a state persistence event. State persistence event INC-2025-084 supports one conclusion and not another: the remediation addresses the immediate cause and leaves the contributing factor in place. The reviewer treats one contributing factor in incident INC-2025-084 as the more important lesson: the pause control had been added in a hotfix without a design review. The affected population in state persistence incident INC-2025-084 was one site and 11 robots. Mitigation of state persistence event INC-2025-084 took 27 m and full resolution took 5 days. state persistence incident INC-2025-084 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Contributing to INC-2025-118, and not repaired by the remediation, is that build logs were retained far longer than the token lifetime, widening the exposure window. Identity and Authorization incident INC-2025-118 belongs to the secret exposure group and sits with Identity and Authorization. Taken with the rest of the record, Identity and Authorization incident INC-2025-118 indicates that the same class of defect can recur in two sibling components that were not changed. Identity and Authorization incident INC-2025-118 was detected because the dependency and secret scanner flagged it on the following scan. Identity and Authorization incident INC-2025-118 was mitigated in 3 h 40 m to revoke and rotate; the resolution ran to 7 days. Identity and Authorization incident INC-2025-118 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The commitment register shows internal incident INC-2025-118 against an undertaking to redact authorization headers at the logging layer and cut build log retention to fourteen days, with Sonia Krall named and 2025-10-31

committed. The affected population in secret exposure event INC-2025-118 was no production site and no robot directly, the exposure being to internal systems. Secret exposure event INC-2025-118 was raised on 2025-09-17 after a service token with a twelve-hour lifetime was found in a build log retained for ninety days. Root cause in incident INC-2025-118 was that a debug logging statement in the connector build printed the full authorization header. On delivery, internal incident INC-2025-118 shows this: redaction landed on 2025-09-19; retention was changed on 2025-09-24. INC-2025-118 would be a candidate for the first post-close engineering review under any reasonable plan.

9. Performance dependence on third-party components

Set beside the other runs, 960-robot run LT-2026-07 varies only in whether the read replicas were enabled. 960-robot run LT-2026-07 is the identity service token storm run, sized at 960 robots and 18,200 orders an hour. Before reading load run LT-2026-07, note that the environment carries production-sized data but a single availability zone. scenario LT-2026-07 was checked against the company's own register, which had not been updated since the change landed.

Before reading 480-robot run LT-2026-08, note that the environment carries a smaller order history than production and a synthetic warehouse endpoint. Load run LT-2026-08 was judged against the fifteen-minute map publication objective. The console session concurrency test, recorded as LT-2026-08, was executed on 2026-02-24. load run LT-2026-08 was checked against the company's own register, which had not been updated since the change landed.

On console session concurrency at twice the plan run LT-2026-09 the reviewer notes that the test environment carried a smaller dataset than production, so the database term is understated here. The constraint surfaced by load run LT-2026-09 is the session relay is single-writer per region. 480-robot run LT-2026-09 was marked fail by the performance team. LT-2026-09 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

LT-2026-10 is recorded as fail against the 1,500 ms p95 acknowledgement target. Reading run LT-2026-10 for the binding constraint gives the outbox table became the limiting term. What run LT-2026-10 does and does not establish is worth stating: the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. run LT-2026-10 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Configuration for load run LT-2026-11 was 240 robots against 8,800 orders an hour in the warehouse connector burst after outbox partitioning case. Load run LT-2026-11 shows where the system runs out of headroom: partitioning moved the limit to the tenant's own endpoint. Configuration of the environment for LT-2026-11 included the current production topology with one region held out. 240-robot run LT-2026-11 rests on a single source, and the reviewer has said so rather than implying corroboration.

LT-2026-12 produced a 1.4 s to console median with 22 s and 48 s tails and an 0.08% error rate. The judgement the reviewer draws from run LT-2026-12 is that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. The target telemetry ingest at twice the current fleet run LT-2026-12 was measured against is the 1,500 ms p95 acknowledgement objective for the warehouse connector. load run LT-2026-12 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The environment behind LT-2026-13 carries a fleet simulator rather than physical robots, with the control loop unchanged, and the reviewer weighted the result accordingly. 1440-robot run LT-2026-13 sits against the 1,500 ms p95 acknowledgement objective for the warehouse connector; whether that target is the right one is a separate question. 1440-robot run LT-2026-13 is not read as a headline number: the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. 1440-robot run LT-2026-13 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-24.

Weighed against the projected fleet, scenario LT-2026-14 indicates that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. LT-2026-14 shows where the system runs out of headroom: drain took 71 minutes and delayed the analytics load past its window. Measurement of 720-robot run LT-2026-14 is against the five-minute configuration distribution objective rather than against a customer commitment. four hour edge buffer drain after an outage run LT-2026-14 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The result recorded for 1000-robot run LT-2026-15 is pass, which the reviewer traced to the underlying measurement. 1000-robot run LT-2026-15 was executed in the performance environment, which carries production-sized data but a single availability zone. Reading map publication to a thousand robots run LT-2026-15 for the binding constraint gives inside the fifteen-minute objective at the tested fleet size. map publication to a thousand robots run LT-2026-15 was cross-checked against the engineering record supplied for TDR-2025-24, and the two agree.

Offline robots applied on reconnect, which is the behaviour the February 2026 incident exposed was the constraint 720-robot run LT-2026-16 exposed. The verdict on scenario LT-2026-16 is pass with observation. In config distribution with thirty per cent of robots offline run LT-2026-16 the median was 1 m 50 s and the tail reached 9 m 20 s; errors ran at 0.00%. LT-2026-16 rests on a single source, and the reviewer has said so rather than implying corroboration.

120-robot run LT-2025-11 returned 210 ms median, 480 ms at the 95th percentile and 760 ms at the 99th, erroring at 0.02%. 120-robot run LT-2025-11 was executed in the performance environment, which carries production data volumes and a stubbed paging provider. Outcome of load run LT-2025-11 was pass, and the reviewer read the raw result rather than the summary. 120-robot run LT-2025-11 was one of two items where TDR-2025-24 could not obtain the underlying data within the engagement.

Before reading 120-robot run LT-2025-12, note that the environment carries a replayed order stream taken from the two largest sites. Single site, peak hour run LT-2025-12 exercised the single site, peak hour case with 120 robots and 2,640 orders an hour, on 2025-10-09. Set beside the other runs, load run LT-2025-12 varies only in the number of sites in the region rather than the order rate. run LT-2025-12 would be a candidate for the first post-close engineering review under any reasonable plan.

LT-2025-13 was judged against the 1,200 ms p95 objective published for order assignment. Single site, peak with a charger outage run LT-2025-13 came out pass with observation. Errors in load run LT-2025-13 were 0.11%, with latency of 310 ms, 870 ms and 1,690 ms across the three percentiles. load run LT-2025-13 rests on a single source, and the reviewer has said so rather than implying corroboration.

Set beside the other runs, load run LT-2025-14 varies only in whether the read replicas were enabled. 240 robots and 4,900 orders an hour define load run LT-2025-14, run on 2025-10-14. Before reading two sites, shared region run LT-2025-14, note that the environment carries the current production topology with one region held out. run LT-2025-14 is included because a reader of TDR-2025-24 should be able to disagree with the inference without repeating the work.

10. Licence questions left open at the review date

Three of the open items in TDR-2025-24 would close on a single document that the company has agreed to produce. M. Quillan has dated each open item in the report at TDR-2025-24 so that a reader can see which have been open longest rather than only how many there are. Stated without qualification for once, the conclusion of the report at TDR-2025-24 is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Calder Rowe Engineering Advisory concludes, on the material examined, that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. The third-party component inventory, licence exposure and contribution provenance of the codebase produced more evidence than the report at TDR-2025-24 reproduces, and the working papers hold the remainder.

Report TDR-2025-24 keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. 4 declined items are carried in the present report, of which two lack any documented compensating control. Interviews for TDR-2025-24 were conducted with the owning engineer present rather than with a manager speaking on their behalf. 21 interviews inform TDR-2025-24. Where an interview is the sole source for a claim, the claim is graded limited. M. Quillan has kept the working papers behind the report at TDR-2025-24 in a form that a second reviewer could re-perform.

TDR-2025-24 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. The report at TDR-2025-24 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. M. Quillan would add one qualitative item to the pack proposed in the present report: the count of commitments closed without verifying evidence. 7 indicators are recommended in this report; the reviewer deliberately avoided proposing anything that needs a new pipeline. Calder

Rowe Engineering Advisory raised each finding with the owning engineer before it was written up, and the responses are recorded.

M. Quillan notes in this report that the company's own reporting is adequate for three of the four recommended indicators. Post-close monitoring under this report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. M. Quillan has left several concerns in TDR-2025-24 at limited evidence strength rather than dropping them or dressing them up. The convention throughout report TDR-2025-24 is that an absence of evidence is reported as such rather than as evidence of absence. The evidence behind the present report is retained by Calder Rowe Engineering Advisory and is available to Ardent Peak on request.

11. Conclusions on intellectual-property hygiene

Whatever weight a reader gives the individual findings, TDR-2025-24 concludes that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. M. Quillan's conclusion in report TDR-2025-24 is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Nothing in this report is presented as verified on the strength of an interview alone. M. Quillan treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. M. Quillan has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

The costed remediation implied by report TDR-2025-24 is $1,030,000 over the coming four quarters, which is within the ordinary engineering budget. M. Quillan would put 4 items from this report into the first hundred days rather than into a warranty. Method in TDR-2025-24 was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. The approach M. Quillan took in report TDR-2025-24 was to test the company's own assertions rather than to restate them. Where TDR-2025-24 uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Monitoring recommended by the present report relies on metrics the company already emits, so no new instrumentation is required to start. The monitoring pack this report proposes would have surfaced two of the incidents examined here before a customer noticed. M. Quillan would summarise report TDR-2025-24 in one sentence: the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. The overall conclusion of this report is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Nothing in this report is intended as a legal opinion or as advice on the adequacy of insurance cover.